

Assignment 1: Implementation of Data Preprocessing Techniques

Objective: Implement the necessary preprocessing steps and prepare the dataset for further analysis. **Dataset:** [UCI Credit Card](#)

Task Overview:

❖ Task 1: Data Understanding and Importing

- Load the dataset into your preferred data analytics tool (e.g., Python, R).
- Print the first 10 rows of the dataset.
- Describe the structure of the dataset (number of rows, columns, types of variables).

❖ Task 2: Handling Missing Data

- Identify any missing values in the dataset.
- Explain the method you will use to handle missing data and justify your choice. ➤

Implement the method and show the result of the dataset after handling missing values. ❖ Task 3:

Data Transformation

- Identify and explain any variables that need transformation or scaling.
- Apply appropriate transformation techniques (e.g., normalization, standardization).
- Visualize the effect of the transformation on one continuous variable.

❖ Task 4: Encoding categorical variables

- Identify categorical variables in the dataset.
- Apply encoding techniques (e.g., one-hot encoding or label encoding) to the categorical variables.
- Explain why you chose this encoding method and how it affects the model's performance.

❖ Task 5: Feature Selection and Engineering

- Remove any redundant or irrelevant features that will not contribute to model building.
- Justify the removal of these features.
- Apply any feature engineering techniques (e.g., creating interaction terms, new variables).

❖ Task 6: Data Splitting

- Split the preprocessed dataset into training and testing sets with an 80:20 ratio.
- Explain why splitting the dataset is essential in the context of model training and evaluation.

❖ **Task 7: Summary of Data Preprocessing**

- Write a brief summary of the data preprocessing steps you implemented.
- Discuss how these steps improve the quality of data for further analysis and modeling.

Note: Comment your code to explain the logic behind each step.